Worksheet 8.6

Science Quest 10: pages 316–17

Newton's Second Law

Answers

1. State Newton's Second Law of Motion.

The acceleration of a body is directly proportional to the net force and inversely proportional to its mass.

2. Use the formula: $F = ma$ to complete the following table

Force (F) (newtons)	Mass (m)	Acceleration (a)
100	40 kg	$\frac{100}{40} = 2.5$ m/s^2
2	200 g	$\frac{2}{0.20} = 10$ m/s^2
70	$\frac{70}{3.5} = 20$ kg	3.5 m/s^2
100	$\frac{100}{0.10} = 1000$ kg	10 cm/s^2

3. Consider the forces acting on a body of mass 40 kg resting on a frictionless surface.



(a) Calculate the net force acting on the body.

$F = 35 - 15 = 20$ N

(b) Calculate the acceleration of the body.

$a = \frac{m}{} = \frac{40}{} = 0.5$ m/s^2

4. The photo shows a drag car coming to a stop at the end of a race.



Photo: Claudia Schnepf / 123RF

(a) To come to a complete stop, the drag car negatively accelerates. Identify another term for 'negative acceleration'. Deceleration

(b) Identify the force that brings the drag car to a stop. Friction

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